JACKSON C. TURNER

QUANTUM RESEARCH ENGINEER | APPLIED MATHEMATICIAN

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Education

Columbia University — **Expected May 2026**
MS/Ph.D. in Applied Mathematics (GPA: 3.97) — *New York, NY*
Engineering Presidential Fellow. Dissertation: Resonance-induced nonlinear bound states and their orbital stability. Advisor: M. I. Weinstein.

University of Utah — **May 2021**
Honors B.S. in Applied Mathematics & Physics (GPA: 3.95) — *Salt Lake City, UT*
Presidential Scholar. Developed mesh-free PDE eigensolvers

Experience

Harmoniqs — Brooklyn, NY
Quantum Research Engineer — 2026 – Present
- Develop the mathematical and computational machinery of quantum control for neutral-atom hardware, focusing on atom transport and optical tweezer control
- Build Julia software for atom control, implementing optimization routines for tweezer trajectories and atom rearrangement

Atlas Fusion — Remote
Mathematical Physicist — Feb – Jun 2026
- Built numerical solvers in Julia for plasma stability problems

Columbia University, APAM Department — New York, NY
Graduate Research Assistant — May 2022 – Present
- Connected quantum scattering theory to nonlinear bifurcation theory, establishing how resonances in linear wave operators generate nonlinear bound states; paper under review at *Nonlinearity*
- Developed perturbative orbital stability criteria for Hamiltonian systems; manuscript in preparation
- Built numerical solvers for scattering, bifurcation, and dynamical systems problems with $< 10^{-8}$ tolerances
- Invited talk, Simons Collaboration on Extreme Wave Phenomena; Best Poster, Joint Alabama–Florida Conference (2025)

Zap Energy, Theory & Modeling Division — Everett, WA
Research Intern — Jun – Aug 2025
- Achieved 720$\times$ speedup (4 hours $\rightarrow$ 20 seconds) building eigenvalue solver in Julia for MHD linear stability analysis; used predictor-corrector continuation with arclength parameterization to track embedded eigenvalues through bifurcations
- Performed spectral analysis of axisymmetric ideal MHD equilibria; studied onset of Kelvin–Helmholtz and shear-driven Mack-mode instabilities via bifurcation theory and complex analysis
- Identified novel plasma instability mechanism in sheared-flow Z-pinch configurations; manuscript in preparation

Volunteer Program Leader — Hong Kong
Full-time Service — 2015 – 2017
- Led teams of 10–20 volunteers across multiple districts, managing goal-setting and training; fluent in Cantonese

Technical Skills

Physics Quantum optimal control, Hamiltonian systems & stability theory, spectral perturbation theory, quantum mechanics, MHD & plasma modeling, eigenvalue problems in complex geometries, bifurcation & parameter continuation methods

Solvers & Algorithms Quantum optimal control (atom transport, optical tweezer control), eigenvalue solvers, parameter continuation and bifurcation tracing (AUTO, BifurcationKit.jl), lattice eigenvalue optimization (flat tori, sphere packing generalizations), spectral methods, ML algorithms

Programming Julia, Python (NumPy, SciPy), MATLAB; familiar with Fortran, PyTorch

Mathematics Spectral theory, scattering theory, bifurcation theory, perturbation methods, variational calculus, constrained optimization, complex analysis, Hamiltonian mechanics

Publications

- **Turner, J. C.** & M. I. Weinstein. "Resonance-induced nonlinear bound states." arXiv:2510.19538, 2026.
- **Turner, J. C.** & M. I. Weinstein. "Lyapunov stability of resonance-induced nonlinear bound states." In preparation, 2026.
- **Turner, J. C.** & D. Crews. "Linear Stability of Ideal MHD in Cylindrical Shear Flows." In preparation, 2026.
- Kao, C.-Y., B. Osting & **J. C. Turner**. "Flat Tori with Large Laplacian Eigenvalues." *SIAM J. Appl. Algebra Geom.*, 2023.
- **Turner, J. C.**, E. Cherkaev & D. Wang. "Expansion Method for Laplace–Beltrami Eigenfunctions." arXiv:2210.10982, 2022.